

TANUSH LAD

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Professional Summary

Structural Engineer with experience in aircraft structural analysis, finite element modelling, CAD design, and engineering calculations for commercial aircraft structures. Performed stress assessment of B777-9 interior assemblies including ceiling panels, class dividers, doorway linings, and structural joints. Proficient in HyperMesh, MSC Nastran, HyperView, SolidWorks, ANSYS, and engineering calculations for structural substantiation. Strong understanding of metallic and composite structures, load transfer, joint design, manufacturing processes, GD&T, and aerospace design documentation.

Education

SRM Institute of Science and Technology, Chennai

B.Tech. Aerospace Engineering

2022 – 2026

CGPA: 8.72 / 10

Technical Skills

Aircraft Structures: Primary & Secondary Structures, Metallic Structures, Composite Structures, Stress Analysis, Load Path Analysis, Joint Analysis, Buckling, Margin of Safety, Sandwich Panels, Structural Substantiation

CAD & Design: SolidWorks, CATIA V5, Part Modelling, Assembly Modelling, Design Detailing, Manufacturing Drawings, GD&T, BOM Preparation, Prototype Development

Finite Element Analysis: HyperMesh, MSC Nastran, HyperView, ANSYS Mechanical, Shell Meshing, Beam Modelling, CBUSH, RBE2, RBE3, CONM2, Static Structural Analysis

Engineering & Manufacturing: Engineering Calculations, Microsoft Excel Automation, Composite Fabrication, CNC Machining, Laser Cutting, Hardware Integration, Static Testing, Validation Testing, Technical Documentation

Experience

Boeing India Pvt. Ltd., Chennai

Aug 2025 – Aug 2026

Stress Engineering Apprentice

- Developed an Excel-based section analysis tool for **15–16 nodes aircraft beam cross-sections** by processing nodal forces and moments to calculate I_{xx} , I_{yy} , I_{xy} , stress distribution, weakest node, and Margin of Safety, reducing manual assessment time by approximately **85%**.
- Built finite element models of **B777-9 interior ceiling assemblies** using HyperMesh through geometry cleanup, midsurface extraction, shell meshing, material assignment, beam property generation, and connection modelling using **CBUSH, RBE2, RBE3, and CONM2** elements.
- Performed structural substantiation of **ceiling panels, class dividers, doorway linings, curtain tracks, oxygen box supports, and lighting assemblies** by evaluating stress distribution, load transfer, structural response, and Margin of Safety under certification load cases.
- Prepared analytical calculations for **lap shear joints, sandwich panels, inserts, brackets, tie rods, and tab-slot joints** using Boeing structural design manuals and D6 documentation to verify structural integrity and design compliance.
- Studied aircraft structural failure modes including **crippling, plastic bending, column buckling, torsional instability, and local instability** for metallic aircraft structures and engineering design validation.

DRDO – Advanced Systems Laboratory, Hyderabad

May 2025 – Jul 2025

Research Intern

- Characterized **EPDM ablative materials** for solid rocket motor insulation through **10+ mechanical and thermal tests**, including density, Shore A hardness, tensile strength, elongation, modulus, rheometry, thermal conductivity, coefficient of thermal expansion, DSC, and FTIR.
- Evaluated thermal protection materials for missile stage applications by analysing material properties, insulation behaviour, manufacturing methodology, and winding configuration.
- Assisted in experimental validation of ablative material performance by comparing mechanical and thermal properties against design requirements for propulsion applications.

Projects

Digital Twin Based Structural Health Monitoring of Aircraft Wing

Software: SolidWorks, ANSYS, Python

- Designed a **500 mm tapered aircraft wing** and integrated **4 strain gauges, 1 load cell, and 1 accelerometer** for real-time structural health monitoring.
- Performed structural sizing, finite element analysis, sensor placement, and experimental validation through bending and vibration testing using an ESP32-based data acquisition system.
- Correlated analytical and experimental structural response to evaluate strain distribution, natural frequency, and structural integrity under static loading.

Rocket Motor Test Rig with Load Cell Instrumentation

Software: SolidWorks

- Designed and manufactured a structural test rig for static rocket motor testing with integrated load-cell instrumentation for thrust measurement.
- Developed the mechanical assembly, structural layout, and hardware mounting arrangement to ensure repeatable testing and reliable force acquisition.

Preliminary Structural Design of a 1 km Altitude Launch Vehicle *Software: SolidWorks, ANSYS, RockSim*

- Designed the primary airframe structure of a small launch vehicle through structural sizing, stability assessment, and manufacturability evaluation.
- Validated structural integrity using finite element analysis while iteratively optimizing component geometry for minimum mass and adequate strength.

RC Aircraft Design and Manufacturing

Software: SolidWorks, XFLR5

- Designed and manufactured a lightweight RC aircraft using carbon-fibre members, Depron foam, laser-cut components, and 3D-printed motor mounts.
- Performed aerodynamic sizing, structural assembly, propulsion integration, and static validation prior to flight testing.

Additional Information

Technical Interests: Aircraft Structural Design, Composite Structures, Metallic Airframe Structures, Structural Certification, Design Detailing, Finite Element Analysis

Leadership: Team SCOUTS – Co-Founder & Mechanical Lead

- Led the structural design and manufacturing of fixed-wing RC aircraft through CAD modelling, composite fabrication, structural assembly, propulsion integration, and flight validation.
- Coordinated a multidisciplinary student team for aircraft design reviews, manufacturing planning, component integration, and experimental testing.

Relevant Coursework: Aircraft Structures, Finite Element Analysis, Strength of Materials, Theory of Elasticity, Composite Structures, Flight Mechanics, Aircraft Design, Manufacturing Technology, Fluid Mechanics, Heat Transfer